Technical Assessment: Data Science

Target Role: Data Scientist (AI & Business Solutions Specialist)

Domain: Commercial Trade, Credit Risk, Sales & Financial Operations

Estimated Effort: ~7–10 Hours of Hands-on Execution

Submission Window: 72 Hours (3 Days)

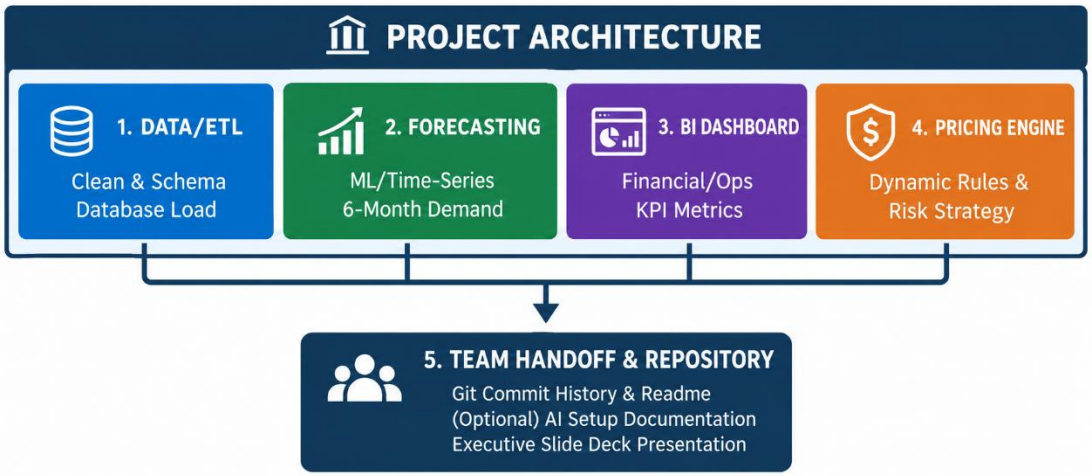
Business Context & Problem Statement

Armani Middle East Trading specializes in the import and distribution of essential food commodities across regional markets. As trade volumes scale, the organization faces key operational challenges:

1. **Demand Volatility:** Supply chain fluctuations, macroeconomic pressures, and seasonality make accurate commodity forecasting difficult.
2. **Dynamic Pricing Needs:** Setting static prices leads to missed revenue during high-demand periods or unsold inventory during off-peak windows.
3. **Credit & Liquidity Management:** Balancing competitive sales terms with exposure to customer credit default risks and inventory holding costs.

As a candidate for the **Data Scientist** role within our Digital Transformation team, your mission is to design and build an end-to-end data pipeline, predictive demand engine, financial KPI reporting layer, and dynamic pricing strategy.

Task Breakdown & Deliverables



1. Data Engineering, ETL & Database Setup

- **Dataset Ingestion:** You are provided with a historical dataset covering **5 years (260 weeks)** of transactional data across **30 essential food commodity products**.
- **Pipeline & Cleaning:** Build a Python preprocessing routine (using Pandas/SQL) to handle missing values, correct schema inconsistencies, remove duplicates, and manage anomalous outliers.
- **Database Modeling:** Ingest the clean dataset into a relational database (PostgreSQL, SQLite, or SQL Server) using a normalized or Star Schema structure (Fact & Dimension tables).
- **Exception Handling:** Ensure your ETL pipeline includes exception handling for corrupt records or missing fields.

2. Time-Series Demand Forecasting & Factor Analysis

- **Predictive Modeling:** Build a machine learning or time-series forecasting model (e.g., Prophet, XGBoost, SARIMAX, or Random Forest) to predict product demand for the **next 6 months (26 weeks)**.
- **Feature Engineering:** Factor in seasonality, trends, historical sales volumes, and pricing signals.
- **Evaluation & Diagnostics:** Measure model performance using standard metrics (MAE, $RMSE$, $MAPE$). Compare your model against a baseline and justify your choice.

3. Financial & Operational KPIs Dashboard

Define, compute, and visualize key performance indicators across three core domains:

Category	Recommended KPIs	Business Objective
Financial	Gross Profit Margin, Revenue Growth, Operating Profit	Evaluate overall commercial profitability.
Operational	Inventory Turnover Ratio, Days Sales of Inventory (DSI)	Optimize supply chain efficiency and warehouse capacity.
Sales & Credit	Customer Default Risk Rating, Days Sales Outstanding (DSO)	Balance sales growth with credit risk exposure.

- **Deliverable:** Develop an interactive dashboard (Power BI, Streamlit, or Tableau) displaying historical trends, 6-month forecasted demand, and credit/inventory risk alerts.

4. Dynamic Pricing & Risk Strategy

Propose an automated or rule-based **Dynamic Pricing Framework** that optimizes profit margins while considering market realities:

- **Demand-Based Adjustments:** How prices scale during forecasted demand peaks or lulls.
- **Inventory & Seasonality Sensitivity:** Pricing rules for high-stock vs. low-stock products during peak and off-peak seasons.
- **Competitive & Credit Risk Guardrails:** Ensuring competitive positioning without eroding profitability or increasing exposure to high-risk credit clients.
- **Deliverable:** A structured documentation section detailing the mathematical rules or algorithmic logic behind your dynamic pricing engine.

5. Repository Structure & Collaboration Standards

As part of our commitment to **Clean Code**, **MLOps**, and **Collaborative Engineering**, your project repository will be evaluated as if you were handing it off to a live engineering team:

- **GitHub Repository:** Host all source code, SQL schemas, and notebook files in a structured GitHub repository.
- **Maintainability & Documentation:** Include a comprehensive `README.md` containing:
  - System architecture diagram or workflow description.
  - Environment setup and installation instructions (e.g., `requirements.txt` or `environment.yml`).
  - Key assumptions made during data processing and model design.
- **Code Quality & Commit History:** Write clean, modular, and well-commented code. Maintain a logical Git commit history (e.g., `feat: add demand forecasting pipeline` rather than a single `initial commit`).
- **Optional — AI Assistant Setup (`.cursorrules` or Prompt Logs):**

  Note: The use of AI coding assistants (such as Cursor, GitHub Copilot, or ChatGPT) is entirely optional and will not grant extra points, higher scoring, or special privileges.

  If you already use these tools in your daily workflow, feel free to include your `.cursorrules` file or custom prompt configurations to show how you manage AI-assisted development.

Dataset Schema Reference

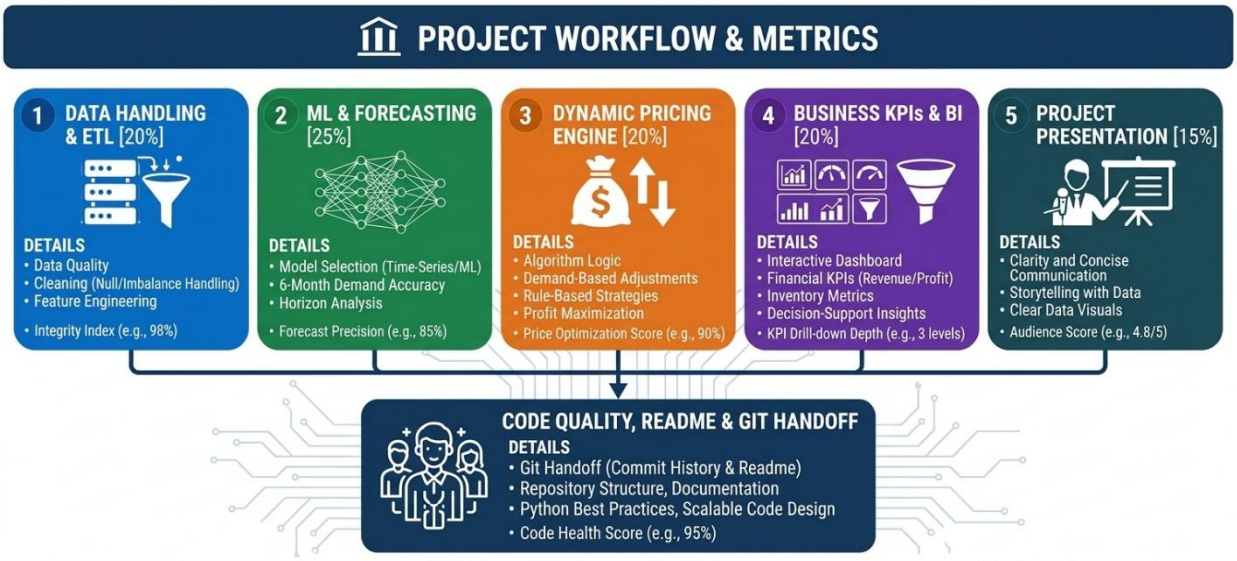
The provided dataset (`Sample_Data.xlsx`) contains the following features:

- `Week`: Sequential week identifier (1 to 260).
- `Product`: Commodity category (e.g., Rice, Wheat, Sugar, Milk, Tea, Coffee).
- `Sales_Units`: Volume of units sold in the given week.
- `Cost_of_Goods_Sold_COGS`: Unit acquisition cost.

- Inventory_Level: Available stock at the beginning of the week.
- Demand_Forecast: Historical baseline forecast (contains missing/noisy records).
- Price_Per_Unit: Historical selling price per unit.
- Gross_Profit: Net margin after subtracting COGS.
- Operating_Profit: Profit after operational expense deductions.
- Revenue: Total weekly sales revenue ($Sales_Units \times Price_Per_Unit$).
- Inventory_Turnover_Ratio: Ratio measuring stock replenishment speed.
- Gross_Profit_Margin: Ratio of gross profit to total revenue.
- Revenue_Growth: Percentage growth compared to previous period.

Evaluation Matrix

Your submission will be evaluated across 5 key pillars:



1. **Data Handling & Pipeline Design (20%):** Robustness of ETL code, database schema quality, and handling of missing or corrupt data.
2. **Forecasting Rigor & Accuracy (25%):** Model selection reasoning, hyperparameter choices, cross-validation metrics, and handling of seasonality.
3. **Business KPIs & Dashboard (20%):** Executive clarity of visuals, relevance of metrics to trade/finance goals, and actionable insights.
4. **Dynamic Pricing Strategy (20%):** Practical applicability of pricing rules, grounding in data, and balance between margin and risk.
5. **Code Quality, Readme & Git Handoff (15%):** Clarity of README.md, meaningful Git commit history, clean repository structure, and code maintainability for team collaboration. *(AI tooling configurations are purely optional and do not impact scoring).*

Required Submission Package

By the deadline, please submit the following:

1. **GitHub Link:** URL to your public or invited private repository containing source code, SQL files, and README.md.
2. **Dashboard File / Link:** Power BI (.pbix) file, Tableau workbook, or live Streamlit web application link.
3. **Executive Presentation Deck:** A concise PDF slide deck (max 8 slides) summarizing:
    - Data pipeline architecture & forecasting approach.
    - Core KPI insights and dashboard walkthrough.
    - Dynamic pricing logic and projected business ROI.